ing to make a faster microkernel. Newer microkernels, though, have tackled the performance issue to a respectable degree.

Off-beat

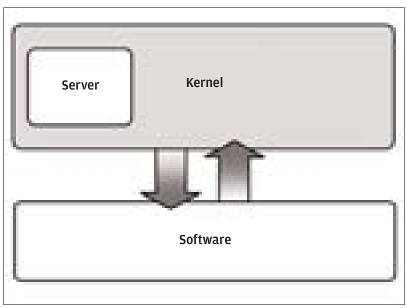
Ununuium (please don't ask us how this is pronounced) is an experimental OS that doesn't use a kernel at all! Instead, it is composed of many components that are loaded into and unloaded from memory as and when needed. This way, no part of the OS is always present in memory, leaving more room for applications to play.

Popular microkernel-based operating systems are Apple's Mac OS X and the Symbian OS. Because crashing servers do not crash the OS itself, microkernels are very stable, and are thus used in applications where the OS cannot be allowed to fail. A notable example is the robotic arms of the Hubble Space Telescope.

1.2.4 The Middle Ground—The Hybrid Kernel

The hybrid kernel is designed to compromise between the monolithic and the microkernel. It still uses servers in the user mode, but it also integrates some of these servers in the kernel mode itself to improve performance. This way, researchers aimed at striking a balance between the speed of the monolithic kernel and the stability of the microkernel.

The hybrid kernel appeared on the scene before it was realised that even pure microkernels could be high performers. Windows 2000 and Windows XP use a hybrid kernel. The microkernel itself is called the *kernel*, and the servers together are called the *NT Executive*.



The Hybrid Kernel is the middle ground between the Monolithic and Microkernel